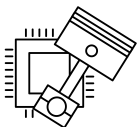
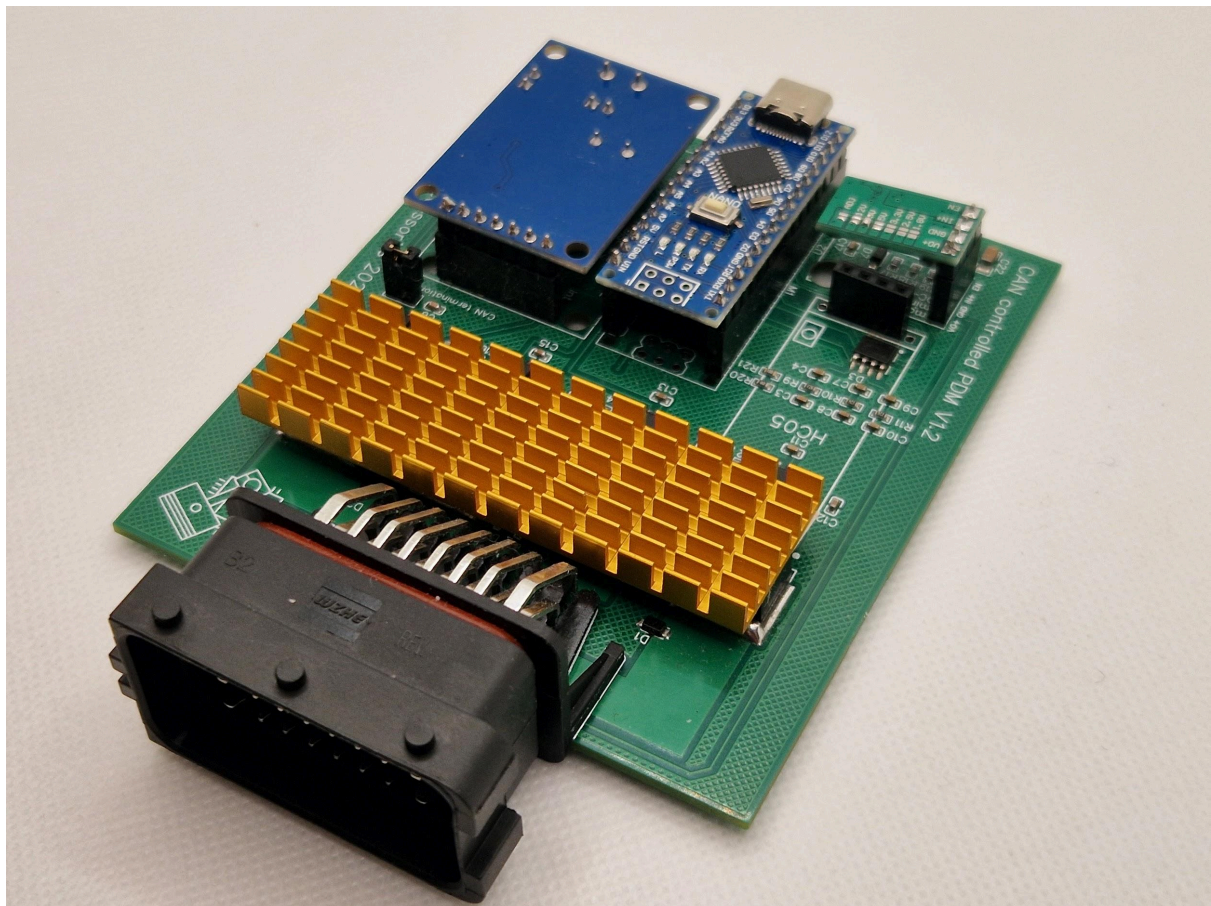
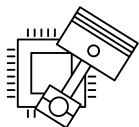


Manual for CAN-Controlled 4-Channel Power Distribution Module



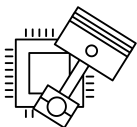
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Introduction

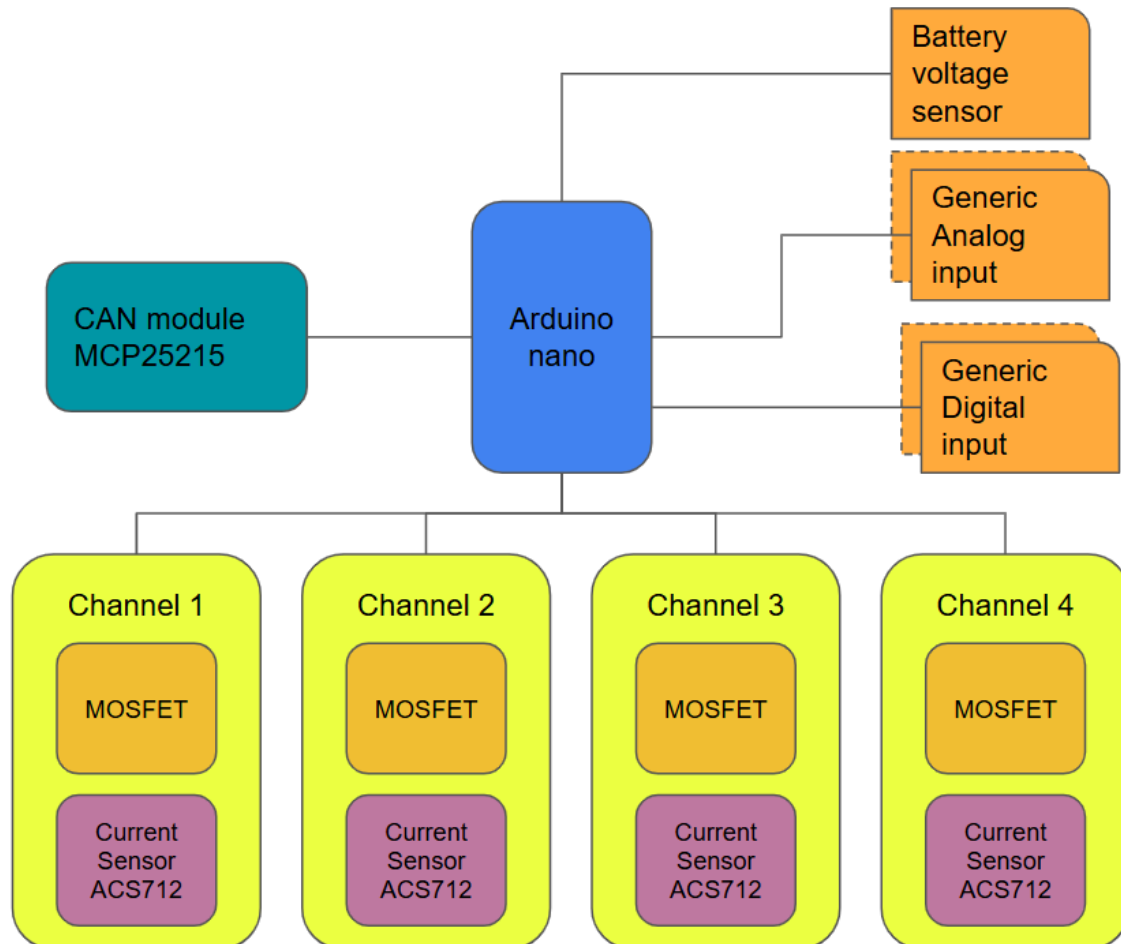


The CAN-Controlled Power Distribution Module (PDM) is an open-source, Arduino-compatible device designed for automotive, industrial, and automation applications. This manual provides installation, configuration, and usage guidelines to ensure optimal performance and integration with your system



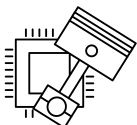
Features & Specifications

Arduino Nano powered PDM



Key Features

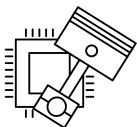
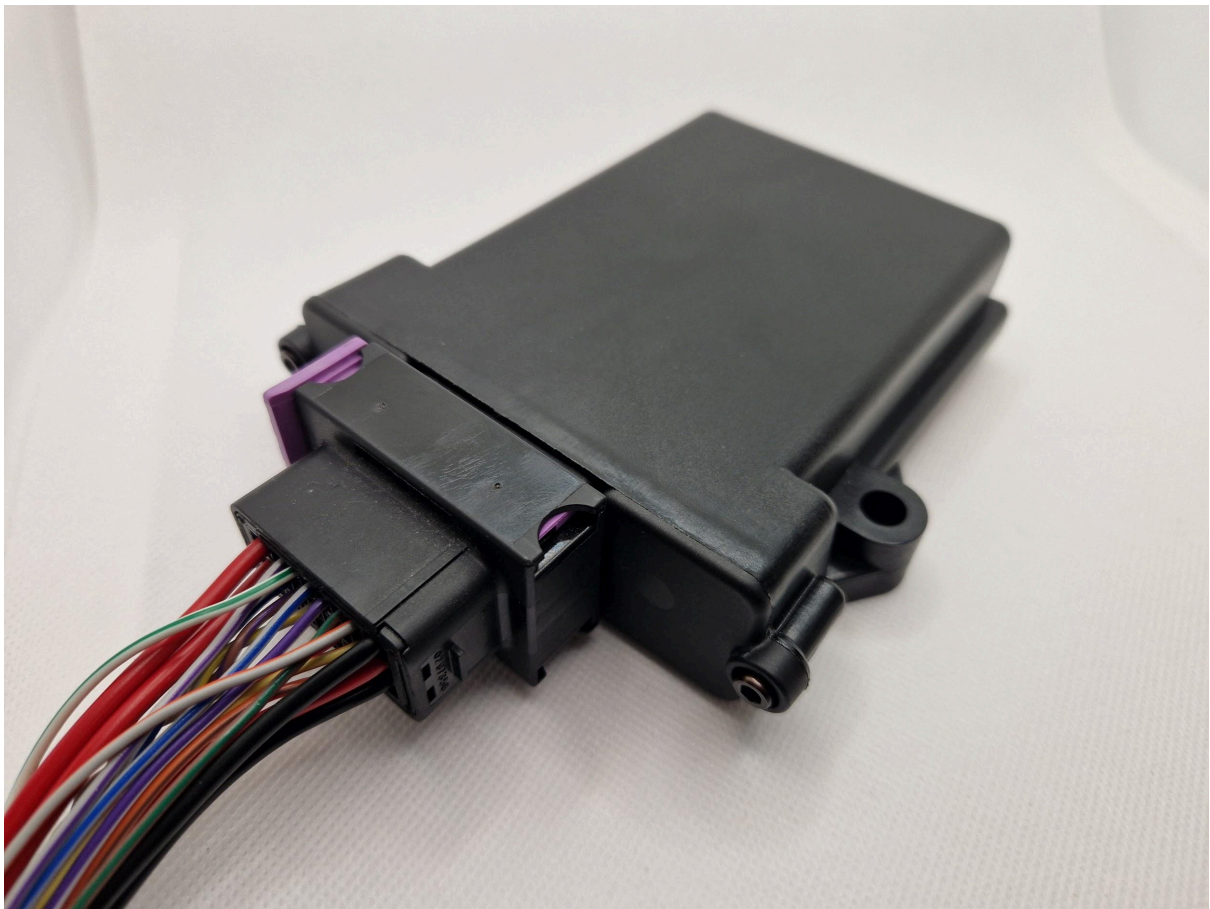
- 4 High-Side Outputs (6A per Channel)
- Current Measurement on Each Channel
- 3 Analog Inputs (0-5V) for Sensors
- 2 Digital Inputs for Switches
- CAN Bus Communication for High-Speed Integration
- Arduino-Compatible for Customization
- Open-Source Hardware & Software
- Selectable CAN Termination via Pin-Header Jumper



Technical Specifications

- **Input Voltage:** DC 7V - 24V
- **Channel Current:** Up to 6A per output
- **CAN Bus Baud Rate:** Default 500 kbit/s (adjustable in software)
- **Power Consumption:** 0.25 W (20mA @ 12VDC)
- **Connector:** 24P FCI
- **Enclosure Dimensions:** 140x120x35mm
- **Microcontroller:** Arduino Nano

Installation & Wiring



Power Supply Connection

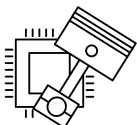
Ensure a stable DC power source within the 7V - 24V range is used. Connect the power input to the designated terminals. 12 volt plus to B1, B8 and C8 Labeled "12V_PWR". Ground to A1 labeled "GND" .

Pinout

The **24-pin FCI connector** is organized into a 3×8 grid (Rows **A**, **B**, **C** and Columns **1–8**), providing the interface towards PDM unit.



Female harness connector.

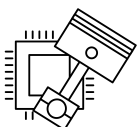




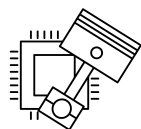
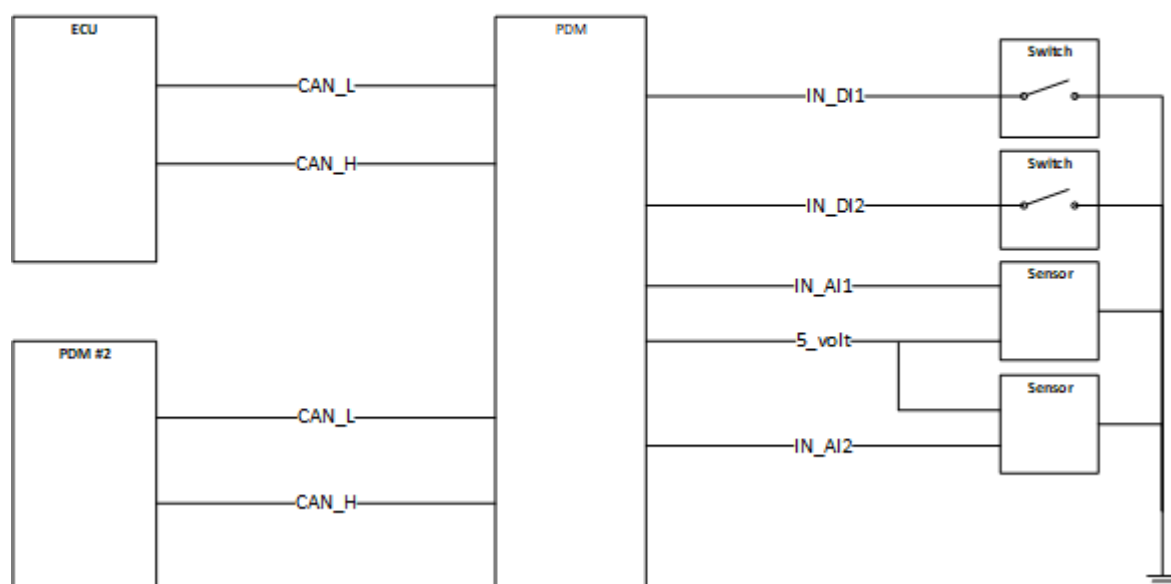
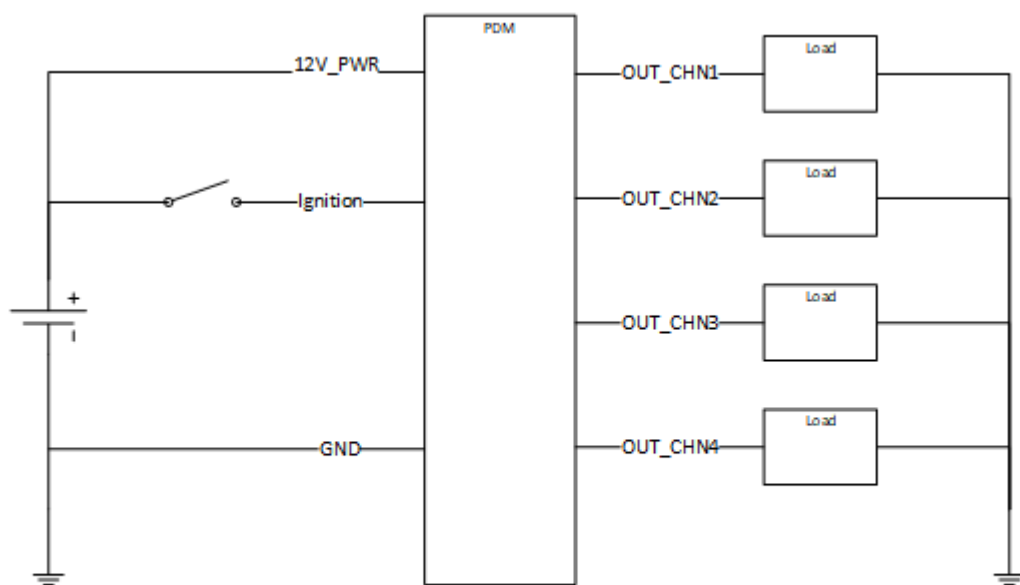
Male connector at the ECU.

Pinout table 24 Pin FCI Connector.

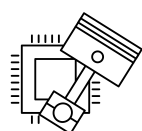
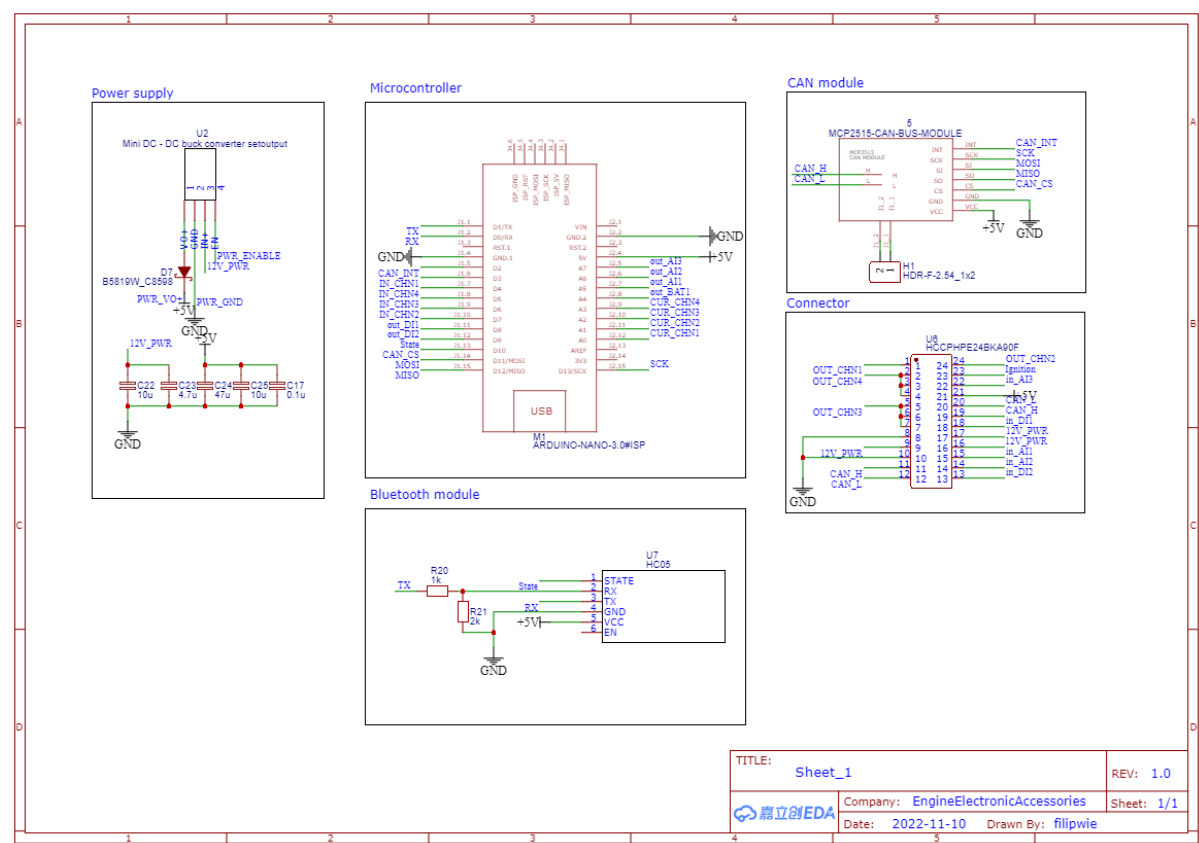
	A	B	C
1	GND	12V_PWR	OUT_CHN2
2	OUT_CHN3	IN_AI1	Ignition
3	OUT_CHN3	IN_AI2	IN_AI3
4	OUT_CHN3	IN_DI2	5_volt
5	OUT_CHN4	CAN_L	CAN_L
6	OUT_CHN4	CAN_H	CAN_H
7	OUT_CHN4	GND	IN_DI1
8	OUT_CHN1	12V_PWR	12V_PWR

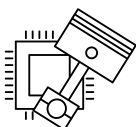
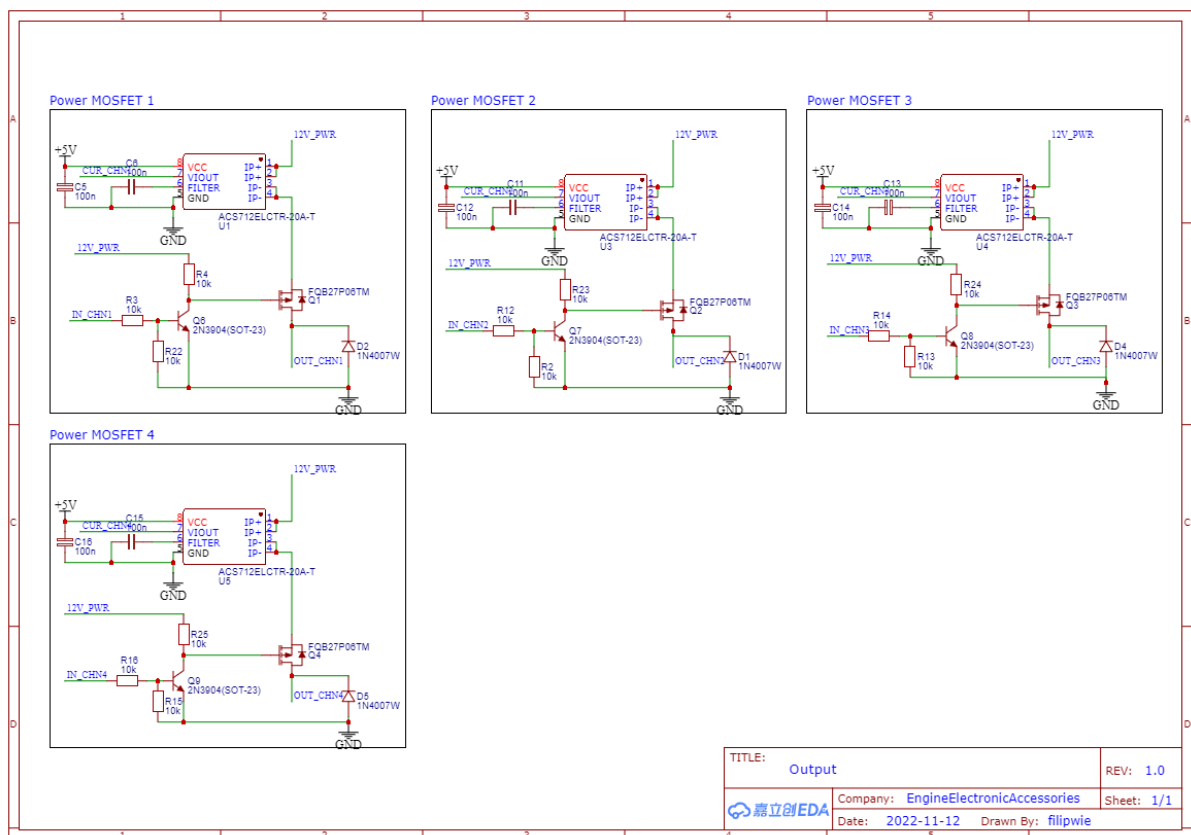
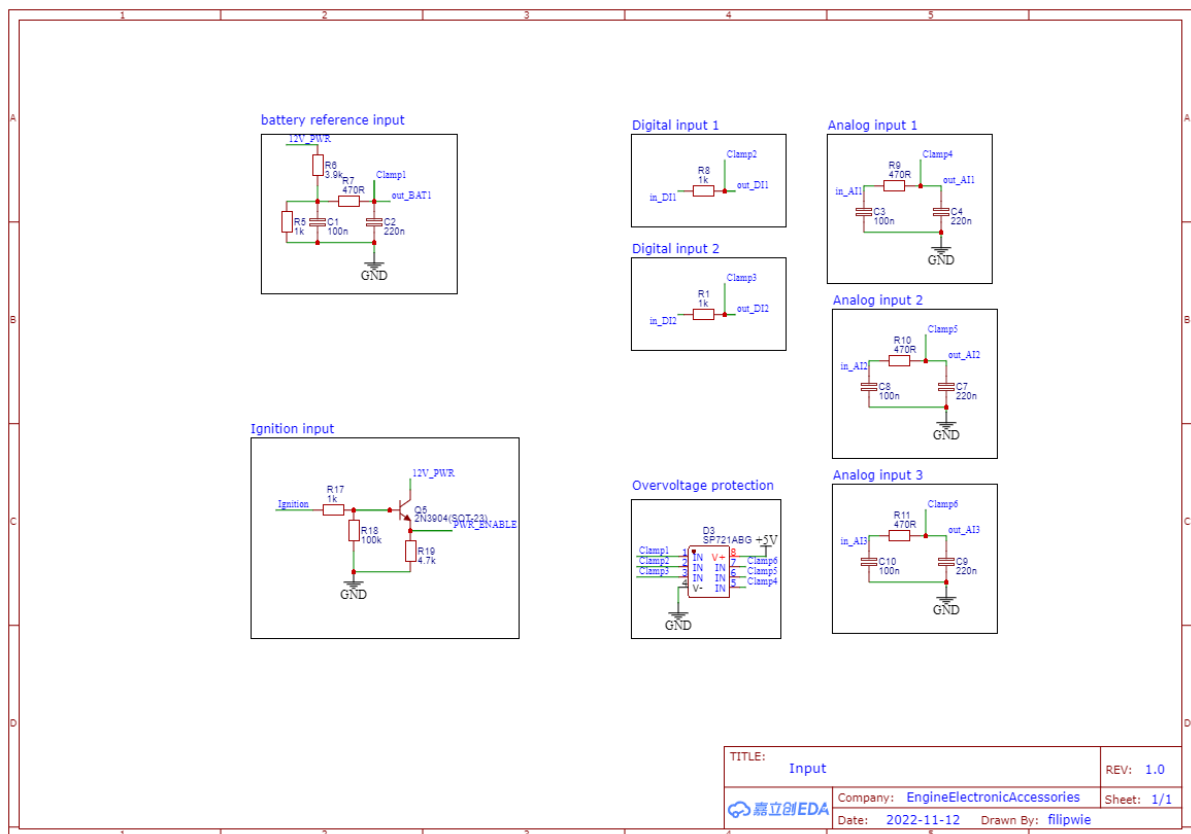


Wiring Diagram



Schematics





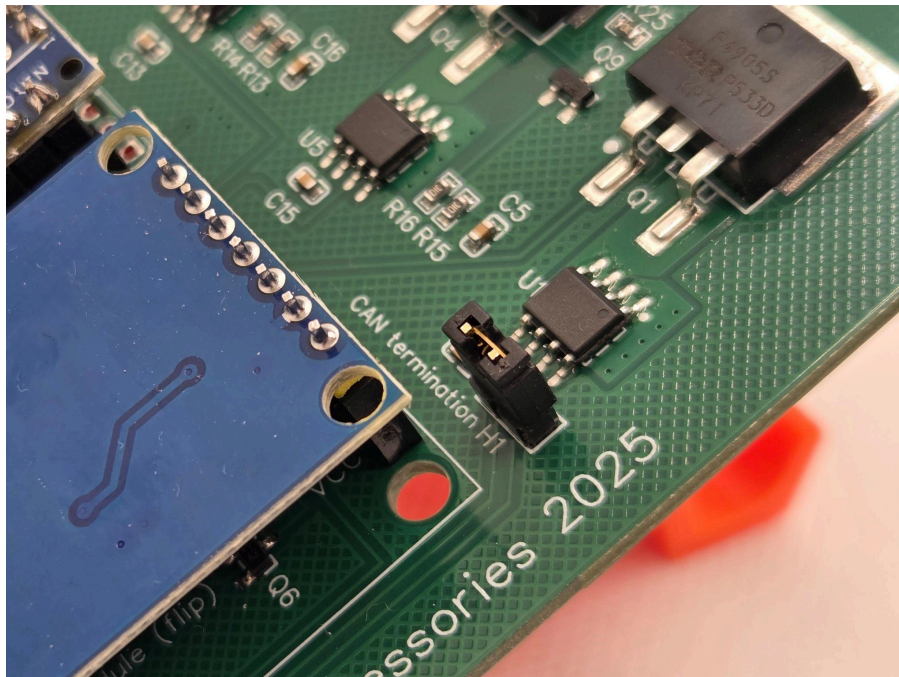
CAN Bus Connection

Connect the CAN High (CANH) and CAN Low (CANL) wires to the respective terminals. If necessary, enable termination using the jumper (120 ohm).

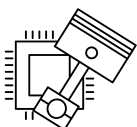
For daisy-chaining, both CAN_H and CAN_L feature dual terminals on the 24-pin connector; these are **electrically identical, straight-through connections**.

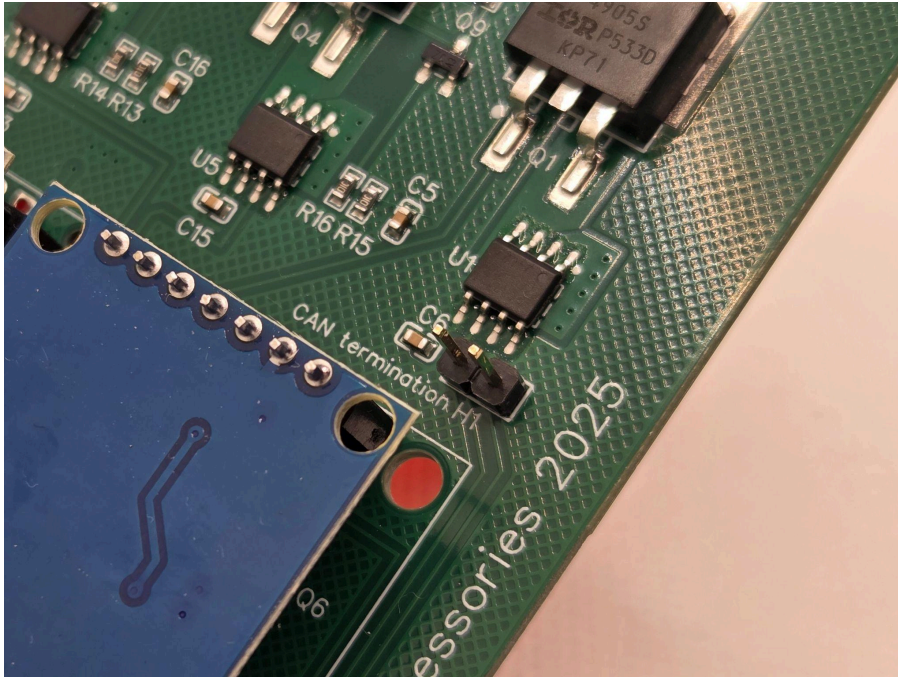
CAN bus termination

CAN bus termination can be toggled on or off using **jumper H1**; installing the jumper enables the internal termination resistor.



CAN bus termination enabled, jumper in place.





CAN bus termination disabled, jumper removed.

Input & Output Connections

- **Outputs:** Connect loads to the high-side outputs, ensuring they do not exceed 6A per channel.
- **Analog Inputs:** Accepts 0-5V signals, ideal for sensors.
- **Digital Inputs:** Suitable for switches and logic signals.

Software Configuration

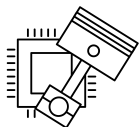
Programming with Arduino

The module is fully compatible with the Arduino IDE. To modify the firmware:

1. Download and install the Arduino IDE.
2. Connect the module via USB.
3. Open the provided source code (<https://github.com/filip-w/PDM>).
4. Adjust CAN settings and logic as needed.
5. Upload the sketch to the Arduino Nano, make shur to select board "Arduino nano" and processor "ATmega328P Old bootloader".

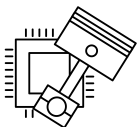
CAN Bus Configuration

Refer to the **PDM.dbc** file for CAN message structure.



CAN baudrate is default set to 500 kbits/s. The speed can be changed by changing the variable “**BaudRate**” in the PDM.ino file.

CAN base ID is 0x500 (1820 decimal) by default. The CAN base ID can be changed by changing the variable “**BaseCanID**” in the PDM.ino file.



Operation & Testing

- 1. Power up the module.
- 2. Verify CAN communication using a compatible tool.
- 3. Send test signals to control outputs and monitor current measurements.
- 4. Integrate with your automation or vehicle system.

Troubleshooting

- **No Power:** Verify the power source and connections.
- **CAN Communication Issues:** Check wiring, baud rate settings, and termination.
- **Output Not Functioning:** Ensure correct wiring, check for overload conditions.

Support & Community

For additional support, firmware updates, and community discussions, visit our open-source repository or contact us.

Link to GitHub Repository:

[filip-w/PDM: Power Distribution Module \(PDM\) CAN controlled Arduino based](#)

Revision

Revision	Description	Date
1	First revsion	2025-03-29
2	Incorrect pinout, Added info about termination resistor	2026-01-14

